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Leuven Case Competition

March 26th, 2026

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Case:
Future proofing AkChem

Case: Future proofing AkChem, a European chemicals player

Our client AkChem, a European chemicals producer, has historically succeeded with reliable, large-scale **fossil-based** production. Today, that model is under pressure: **European energy and CO_2 costs have risen structurally**, and **global overcapacity is depressing margins** across the value chain

At the same time, **customer demand is shifting toward sustainable chemicals**, driven by major brand commitments and tightening regulation

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To stay competitive, AkChem must **reposition its portfolio** and **secure access to sustainable feedstocks and routes**—such as recycling-based inputs, biomass, or CO_2-derived pathways—while moving fast enough to build credibility on scale, traceability, and emissions performance

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The board has asked the CEO to develop a concrete plan to navigate possibilities for this transition and set-up the company for future success. The CEO has asked your team for help to **assess key considerations, identify the most attractive sustainable segment to transition to, and determine for which sites the shift is financially viable**

Europe is facing a plastic waste challenge…

+450%

Increase in end-of-life plastics between 1970-2020 with projection of further increase until 2030 by ~15 MTA

4-6%

Of total global GHG emissions are today driven by plastics production and incineration at end-of-life – 70% reduction by 2030 required to meet 1.5°C trajectory

… but demand for sustainable plastics is accelerating

50%

Minimum recycled content on plastic packaging mandated by the European Union –

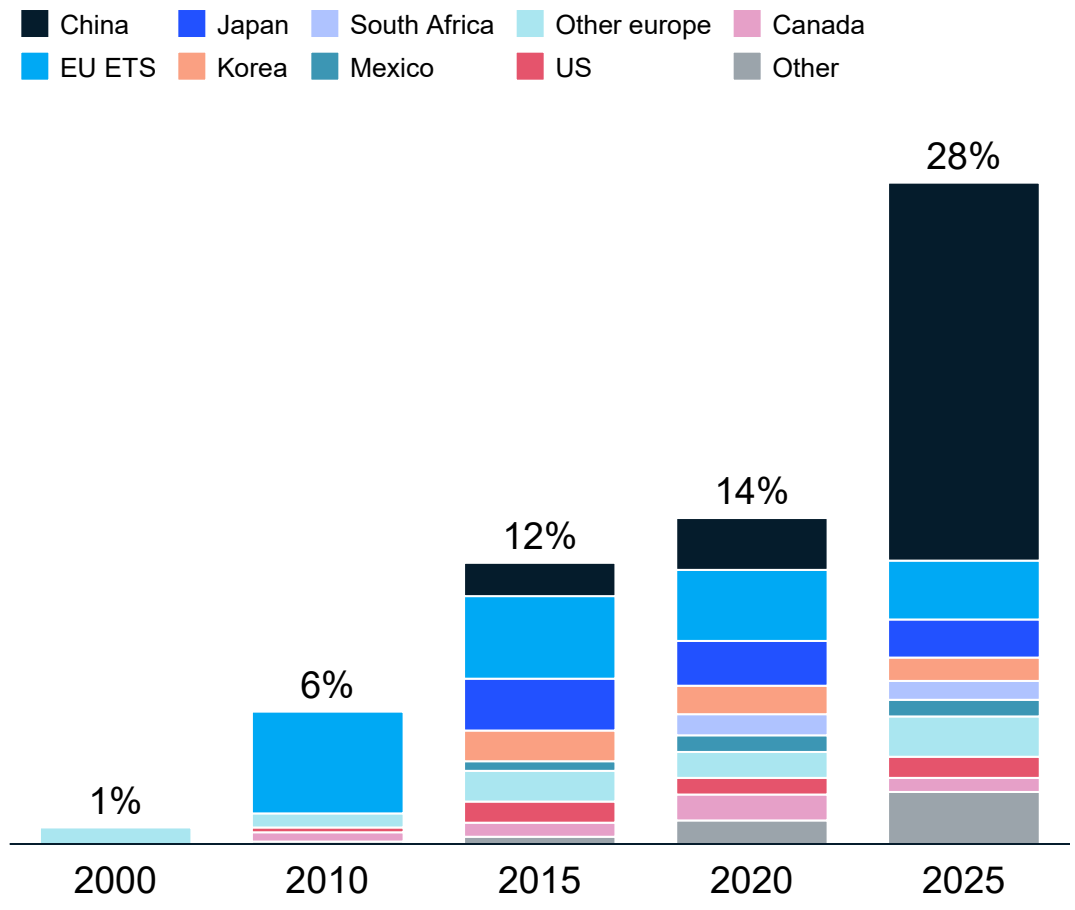
additional voluntary commitments on recycled content set by leading brand-owners and retailers

66%

Of companies in relevant plastic use segments have committed to ambitious medium-term GHG emission reduction targets, driving demand for low-carbon plastics

Carbon pricing is expanding through major economies…

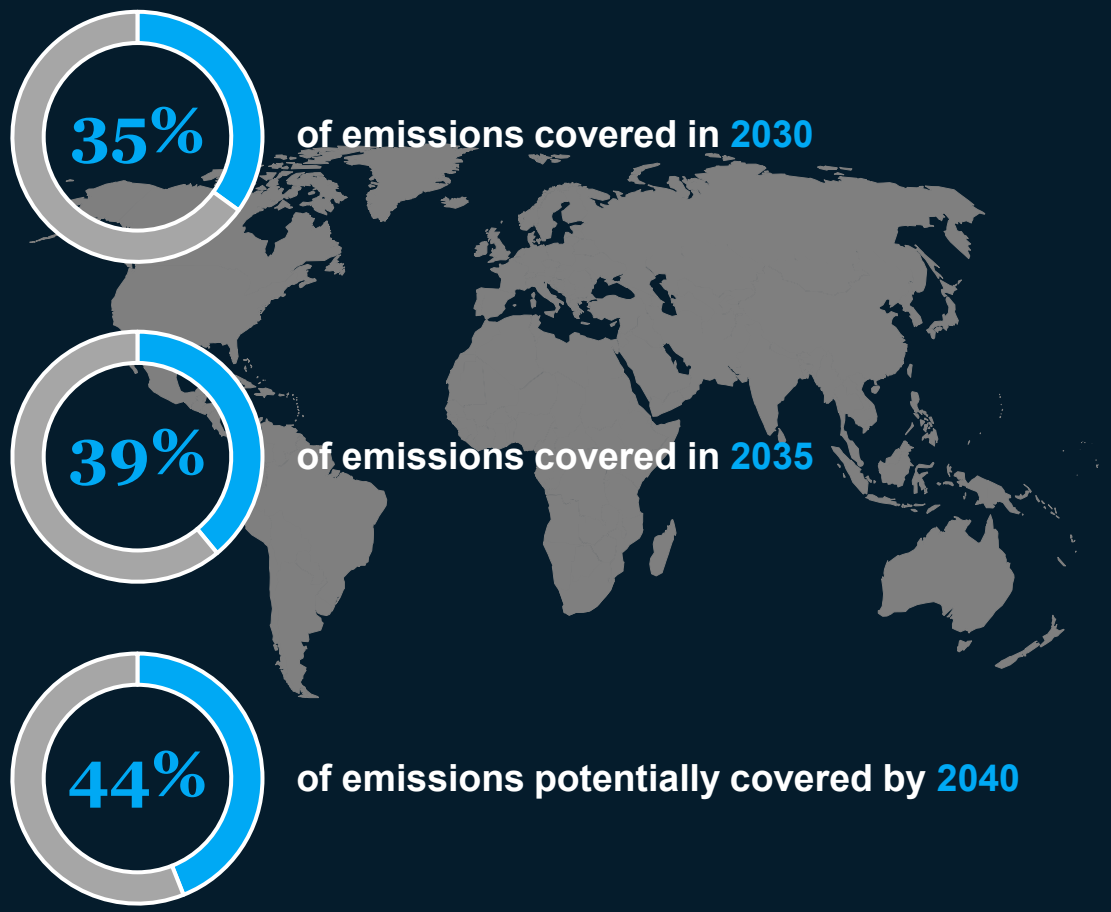
Share of global GHG emissions covered by carbon pricing policies, in %



Source: World Bank Carbon Pricing Dashboard, July 2025

…and may cover ~40% of global emissions by 2035

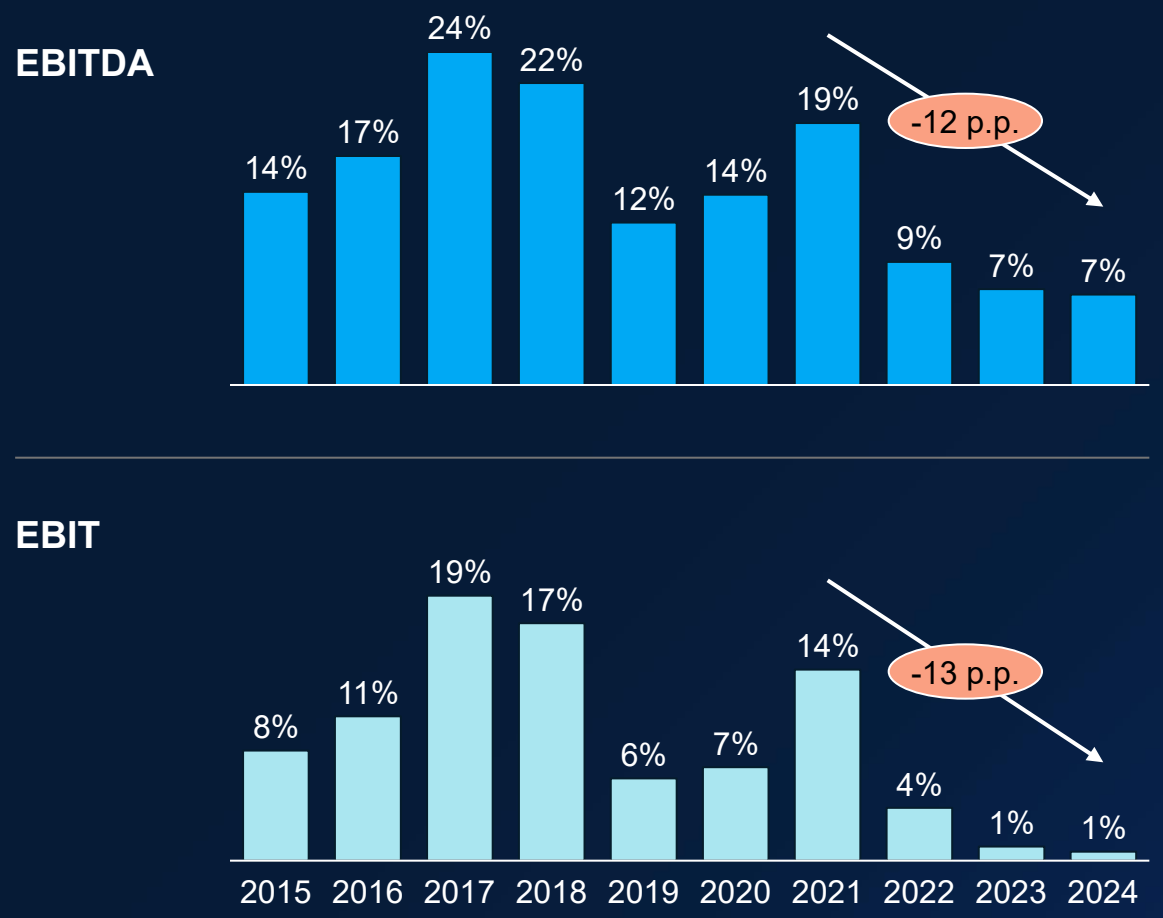
Proportion of global emissions covered by carbon pricing



Source: McKinsey's Carbon Markets Solution

AkChem has faced decreasing EBITDA and EBIT margins over the past years

Margin evolution for AkChem, in %



Source: Company information

Key information and figures

Business focus
Specialty and commodity chemicals

2024 revenue
10.2 bn EUR

Number of employees
18,000

Channels
Mixed (direct to customer + distributors)

Legacy
>80 years of experience

Governance
HQ: Cologne, Germany

How can AkChem become future proof?

The CEO of AkChem expects your team to answer the following questions:

1. What **elements** should the company consider when moving to **sustainable chemicals**?

2. Which **sustainable pathway** should AkChem prioritize?

3. Should AkChem **close any fossil-based sites,** and does **switching to sustainable pathways** make **financial** sense**?**

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You will get various inputs to build your case

Current trends in Europe's chemicals industry

Perspective on sustainable chemicals

Client information and communication with management

Market trends | Market demand for sustainable materials continues to gain momentum from government, companies, and consumers

7.5% of global GDP estimated to go into green businesses investments to reach net-zero by 2050

80% of investments flowing into carbon-intense sectors (incl. chemicals and materials)

European Union's Packaging and Packaging Waste Regulation requires **55%** of plastics to be recyclable by 2030

~66% of companies in relevant plastic use segments have committed to ambitious medium-term GHG emission reduction targets

CPG[1] brands use of recycled plastics increased by **3x** between 2018 and 2021

European Commission has set the target that by 2030 sustainable non-fossil sources should cover at least **20%** of carbon use for chemical and plastic products

1. Consumer Packaged Goods

Source: Preqin, Ellen MacArther Foundation, CapIQ, Science-based targets initiative, CDP Worldwide, McKinsey ESG Solutions / Sustainability Insights, U.S. Consumer Survey, European Parliament (2022)

Market trends | EU packaging rules create mandatory demand for recycled content (2030)

Driven by regulation

"By 1 January 2030 or 3 years after the date of entry into force of the implementing act referred to in paragraph 8, whichever is the latest, any plastic part of the packaging placed on the market shall contain the following minimum percentage of recycled content recovered from post-consumer plastic waste, per packaging type and format, calculated as an average per manufacturing plant and year"

Packaging type		Minimum required recycled content
1 PETE	**Contact sensitive packaging** made primarily of PET[1]	2030: 30%; 2040: 50%
5 PP	**Contact sensitive packaging** (plastics other than PET)	2030: 10%; 2040: 25%
	Disposable plastic beverage bottles	2030: 30%; 2040: 65%
7 Other	**Other plastic packaging**	2030: 35%; 2040: 65%

1. Except single-use beverage bottles

Source: EU Packaging & Packaging Waste Regulation (PPWR) – minimum recycled content provisions.

Market trends | High prices for natural gas and electricity are resulting in high production costs for energy-intensive chemicals

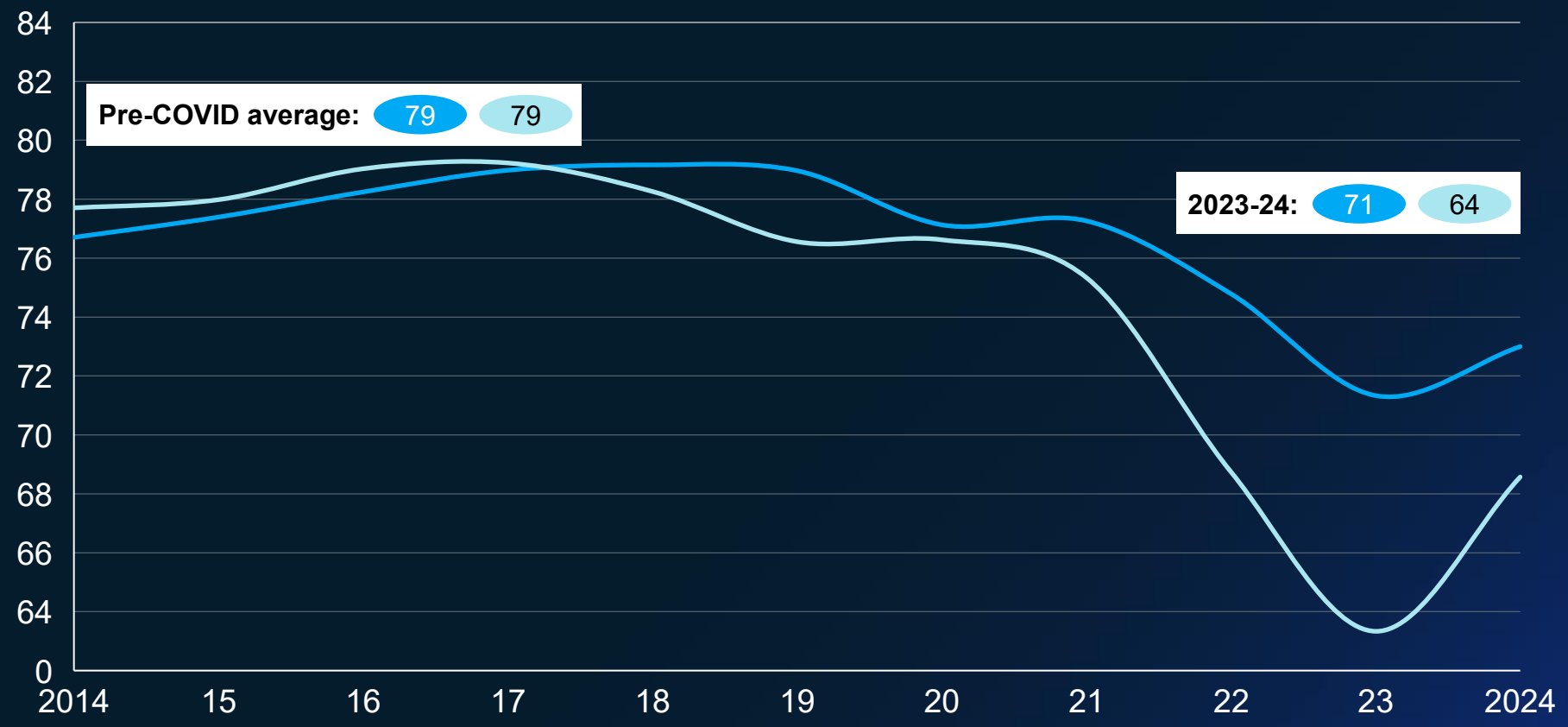
1. Million British Thermal Units, a unit of energy typically used to measure large amounts of gas and energy

Source: Bloomberg, CEM

Market trends | Global overcapacity is depressing margins and volumes, with Europe disproportionately impacted

XX Average utilization — Global — Western Europe

Global vs. Western Europe petrochemical utilization rate[1], %



Supply-demand imbalance resulted in **low-cost producers continuing to produce,** pressuring utilization and margins in high-cost regions like Europe

1. 2EH, acetic acid, acetone, acrylic acid glacial, CAN, ABS, adipic acid, AND, aniline, BZ, BPA, BDE CPL, CHX, cumene, EB, ethylene, EDC, EO, HMDA, isophthalic acid, methanol, MMA, MDI, NBR, PA6/66, phenol, PBR, PBT, PC, polyether polyol, PE, PP, PET, PS, PVC, PO, propylene, SBR, SM SAP, PTA, toluene, TDI, VAM, VCM, MX

Source: Chemical Market Analytics by OPIS

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Sustainable pathways | Overview of the shift towards sustainable chemicals

Sustainable Chemicals: From Emerging Concept to Industrial Reality

The chemical industry is under growing pressure to embrace greener, low-carbon alternatives. What are sustainable chemicals, how do they work, and what are the risks?

What are sustainable chemicals?

Sustainable chemicals are products that are produced with significantly lower environmental impact and carbon footprint over their entire life cycle. Alternative Carbon Sources, instead of relying solely on oil and gas, they use **recycled waste, biomass** (e.g., plant residues, vegetable oils), or **captured carbon dioxide** (CO_2) as raw materials. The idea is to keep carbon circulating in the economy rather than releasing it into the atmosphere.

How do they work?

There are three key technological pathways for sustainable chemical production:

- The sector is shifting from oil and gas to alternative feedstocks
- **Recycling-based**, **bio-based**, and **CO_2 - based** routes are the main-pillars
- Bio-based chemicals account for less than 10% of production
- CO_2 derived chemicals remain largely in the pilot stage
- Investments are rising, but risks are still present

How do they work?

There are **three key technological pathways** for sustainable chemical production:

- **Recycling-based feedstocks** (closing the loop): Chemical-or “advanced" recycling processes break plastic waste down to its basic molecules so they can be reused as raw materials for new chemicals or plastics. This means waste is incinerated or landfilled.
- **Bio-based chemistry:** Uses renewable biological materials such as plant residues and vegetable oils as feedstocks for chemical production. Through fermentation or chemical conversion, plant-based carbon is transformed into chemicals that can replace fossil-based products
- **CO_2 - based production** (carbon capture and utilization): CO_2 emissions from industrial processes – or even from the air itself – are captured and chemically converted into useful products like methanol or polymers, often using renewable electricity and green hydrogen.

How far developed is the technology?

Sustainable chemicals are still a small fraction of total production globally. **Bio-based chemicals make up less than 10%, chemically recycled plastics represent under 1%** and **CO_2 - derived chemicals largely exist at pilot** or demonstration

- Investment is **growing rapidly**, with **Europe leading** the way. Billions of euros are being poured into recycling plants, biorefineries, and carbon utilization projects.
- Many major chemical firms now offer certified "circular" or "bio-attributed" products using mass-balance systems that mix sustainable and fossil-based feedstocks in existing facilities.

Source: Press search

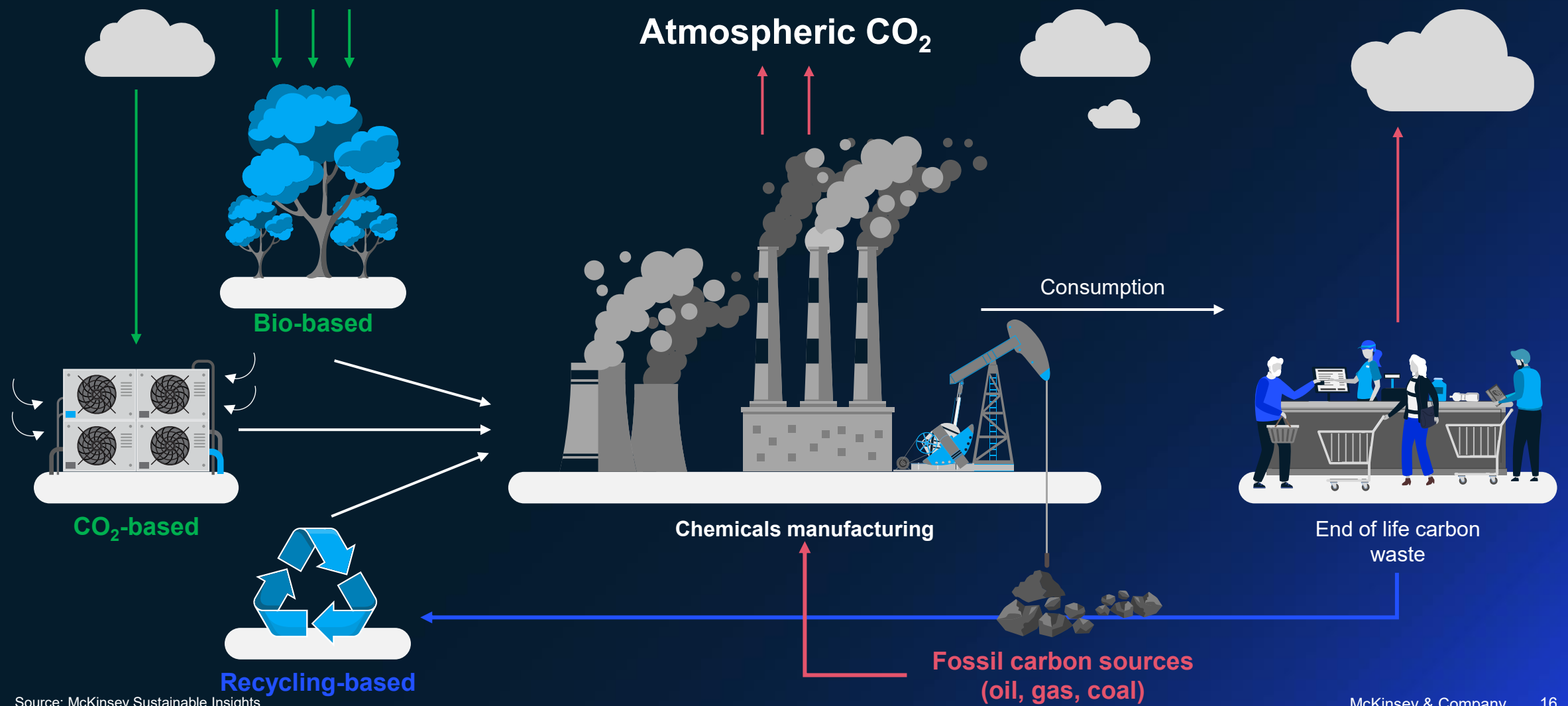
Sustainable pathways | Different sustainable feedstocks and pathways exist within chemical manufacturing

Schematic illustration of CO_2 sources

CO_2 and bio-based (e.g., DAC/Biomass) take CO_2 out of the atmosphere

Recycling-based avoid CO_2 emissions

Fossil: add CO_2 to the atmosphere



Source: McKinsey Sustainable Insights

Sustainable pathways | The different sustainable pathways vary in their value propositions to key customers

Indicative

Value proposition: High Medium Low

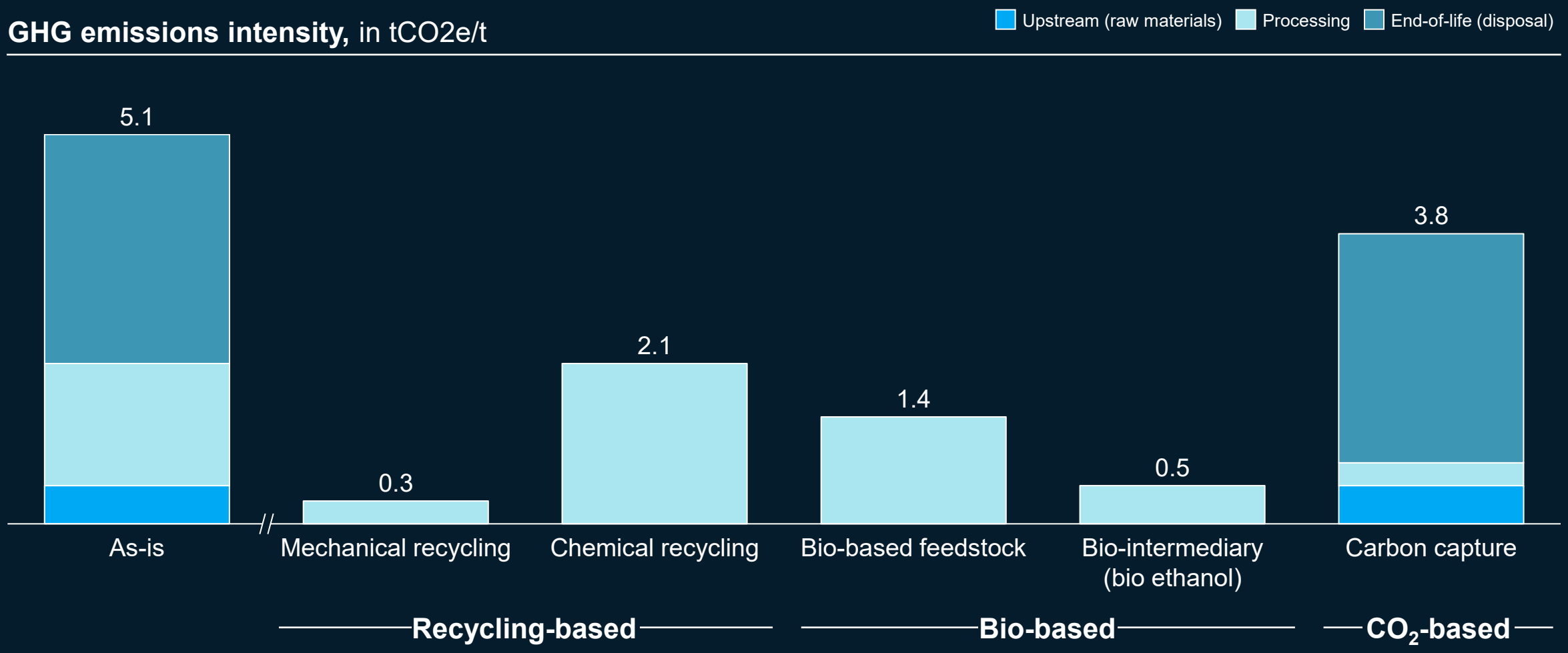
Value proposition	Recycling-based		Bio-based		CO_2-based
	Mechanical recycling	Chemical recycling	Bio-based feedstock	Bio-intermediary	Carbon capture
CO_2 reduction potential	High	High	High	High	Medium
Circularity potential[1]	High	High	Low	Low	Low
Quality potential	Medium	High	High	High	High
Cost competitiveness	High	Medium	Medium	Low	Low

1. High score if the technique actively contributes to transform plastic waste into recyclate reducing the need for fossil plastics

Source: McKinsey Sustainable Insights

Sustainable pathways | GHG reduction potential vary across sustainable plastic pathways and feedstocks

Indicative for high-density polyethylene



Source: McKinsey Sustainable Insights

Sustainable pathways | Availability of feedstocks can impact the choice of the preferred technological green plastic route

Not exhaustive

High | Medium | Low

Type	Pathway	Demand perspective	Supply perspective	Availability
Recycling-based	Mechanical recycling	**Strong demand** driven by circularity targets and regulation **Widely adopted**, but limited to lower-quality / non-food applications	**Requires efficient waste collection** and sorting systems **Constrained by logistics, contamination** and inconsistent input quality	Medium
	Chemical recycling	**Increased demand for high-quality** and food-grade recycled plastics	**Depends on the segregation of single polymers** which is challenging in mixed waste streams especially with compound materials	Low–Medium
Bio-based	Bio-based feedstock	**High demand due to drop-in compatibility** with existing assets	**Depends on agricultural feedstocks**, with sugarcane commercially available and widely cultivated, and corn sugar readily accessible **Constrained by competition** with food supply **Limited availability of waste** vegetable oil	Medium–High
	Bio-intermediary (bio ethanol)	**Increased demand driven by shipping industry**, not yet used for plastics **Prohibitively expensive** for plastics without mandates	**Highly available** and potentially unlimited with direct air capture and solar/wind power	High
CO_2-based	Carbon capture	**Emerging demand driven by decarbonization targets** **Strong interest from aviation, shipping, and chemicals**, but adoption still nascent	**Depends on availability** of CO_2 and **low-cost green hydrogen** (i.e., often considered as key constraint given limited supply in Europe)	Low–Medium

Source: McKinsey Petrochemicals Practice

Sustainable pathways | Technology maturity and commercial scale differ across sustainable plastic pathways

Indicative Not exhaustive

High Medium Low

Type	Pathway	Tech maturity	Rationale
Recycling-based	Mechanical recycling	High	**Commercially established and widely deployed** at scale, leveraging proven infrastructure **Limited by input quality**, particularly for high-specification applications
	Chemical recycling	Medium	**Select technologies** (e.g., pyrolysis) are **reaching early commercialization** with growing adoption but remains energy-intensive and costly to scale **Gasification and solvent-based purification remain at early or prototype stage**, with high capital intensity and scalability constraints
Bio-based	Bio-based feedstock	Medium	**Multiple routes in development** with different feedstocks (e.g., wood and sugar)
	Bio-intermediary (bio ethanol)	High	**Mature, industrialized** conversion pathways deliver consistent performance at scale
CO_2-based	Carbon capture	Low	**Technologies remain largely pre-commercial** with limited large-scale deployment to date **Dependent on advancements in renewable energy** and supportive policies

Source: McKinsey Sustainable Insights

Sustainable pathways | Sustainable routes have different growth perspectives and CapEx intensities

High ● Low ● **Growth perspective:** Very high ↑ ↗ → ↘ ↓ Very low

Type	Pathway	Growth perspective	CapEx intensity	Time to market
Recycling-based	Mechanical recycling	↗		1–3 yrs
	Chemical recycling	↗		2–8 yrs
Bio-based	Bio-based feedstock	↑		2–8 yrs
	Bio-intermediary (bio ethanol)	↑		2–8 yrs
CO_2-based	Carbon capture	↑		5–10 yrs

1. Shows the degree of overlap with the current fossil fuel chemicals

Source: Market research (Grand View Research; Global Market Insights) and public outlooks; 2023/24 baseline, indicative ranges

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Perspective on sustainable chemicals

Client information and communication with management

Communication | An email from the CFO

Subject: Confidential – funding options

From: CFO@Akchem.com

To: CFO@Akchem.com

Hi Team,

Happy to have your support. This project has been giving me quite a headache the past days, since it is also the deadline for our quarterly, financial update. With the management team we had been discussing the possibility of moving to sustainable chemicals now for a couple months, but while others focus on the strategy, I have a problem with high investment costs. Given the **low margins and overcapacity** on the market, we are thinking of **updating one of our current sites** to be able to deal with sustainable chemicals. Our head engineer, Janit, mentioned they had figured out a way to switch around the **base process from fossil based to sustainable products**. You should reach out to him. I know that when producing sustainable chemicals, our **revenue is expected to grow yearly 4% as opposed to 1% growth without change**. Internally, we look at all our projects with **2050 as an endpoint**.

Please explain to me if **closing and switching around one of these sites would make sense** from a financial standpoint. Btw, I cannot disclose much yet but for our **2026 financials**, I propose you take an **average of the past 3 years.**

Good luck, looking forward to your proposals!

Sarah
CFO
AkChem

Communication | A consultant's chat history with the Lead Engineer

Hi Janit,

Hope you are well. Sarah told us you might have some extra data specifics regarding the maintenance and CapEx of your three sites. Shall we schedule a call with you?

Hi team,

Yep, she told me. Currently not much time though. What you need to know is the following. There are in my opinion two options. We switch to sustainable or keep on producing like before

Option 1: The switch would mean an **investment of EUR 250 m** in 2026 followed by a yearly **4.5% CapEx** (in % revenue) spent from **2027 onwards**. It has a useful life till 2050

Option 2: If we stay producing like before, we do not make the investment but we still have the **yearly 4.5% CapEx** (in % revenue) **spent starting 2026**

Any information you have on depreciation of machinery or replacement that need to happen quickly?

Nothing special, but I remember our CFO once mentioning how funny it is our depreciation always matches the changes in net working capital, so we do not have to include it in a free cash flow calculation. Also, I have the best data on the following 3 sites: Antwerp, Hamburg and Barcelona

Sure, thanks!

Financials | AkChem has 3 sites around Europe pre-selected for a potential switch to sustainable chemicals

Detailed input on AkChem's sites financials

2023 | 2024 | 2025 | Fossil based | Sustainable chemicals

Location	Revenue, kEUR	Tax rate	WACC (discount rate)	EBIT margin, %, industry benchmark
Antwerp	560,095 536,721 510,134	30%	7.5%	7% 8%
Hamburg	651,174 686,502 623,760	27%	6.3%	4% 9%
Barcelona	231,056 273,636 256,134	25%	8.3%	1% 10%

Source: Company information

Communication | Auditor input: site closure costs & green subsidies

Subject: Confidential – site closure costs & green subsidies

From: audit.team@auditfirm.com

To: project.team@akchem.com

Hi team,

Sharing a few inputs you can use in your business case:

Closure / switching costs (if you close a site): plan for one-off cash costs across (1) severance/social plan, (2) contract termination, (3) decommissioning & dismantling, and (4) environmental remediation. In most cases, cash-out is spread over ~2–4 years, with remediation as the biggest uncertainty.

Green support: for sustainable-chemicals investments in Europe, support typically comes as (a) CapEx grants, (b) OpEx support (e.g., carbon contracts-for-difference / operating aid) and/or (c) preferential green loans/guarantees. Treat subsidies as uncertain until confirmed

Hope this helps.

Best regards,

Sustainable projects regularly face roadblocks

Some players have delayed their scale-up due to various cited reasons (e.g., technical issues) …

Skive plant (20 kta)

*"... planned to be set into production in **2019"** (Annual Report 2018)*

Operational and producing pyrolysis feedstock; focus on ramp-up and efficiency improvements. Ongoing pipeline expansion under evaluation (2025)

Ashley plant (100 kta)

"... today [May, 2019] broke ground on the nation's first commercial-scale plastics-to-fuel plant in Ashley, Indiana" (Brightmark website)

The facility faced financial distress and bankruptcy of associated subsidiaries in 2025. Brightmark's parent company acquired the plant to keep it operating, though it has historically operated at low capacity (~5%)

Teesside plant (4*20 kta)

"... first 20 kta line having begun construction in April 2021, due to be operational by late 2022" (MURA Brochure 2022)

"... will commence operations in Q1 2026, as the first commercial-scale advanced recycling facility in the UK with a 20kta capacity..." (2025)

Source: Quanafuel Annual Reports, Brightmark webpages, MURA technologies Brochure, Reuters, PlasticsEurope, Pryme update Oct 2023

… While others have announced project cancellations due to financing, weak market conditions, and regulatory challenges

Chemical Recycling Project

- **Location:** Antwerp, Belgium; Rotterdam, Netherland
- **Capacity:** 80kta
- **Reason:** Concern over EU's draft rules for calculating the amount of recycled content in a final product
- **Announcement:** September 2025

IRG Plastic Recycling Project

- **Location:** Erie, Pennsylvania, USA
- **Capacity:** 100kta
- **Reason:** Failure to rise the financing needed for the project
- **Announcement:** April 2025

Mechanical Recycling Plant

- **Location:** Schwechat, Austria
- **Capacity:** 60kta
- **Reason:** Facility would not meet expected performance targets under current market conditions
- **Announcement:** August 2025

Source: Recycling product news, Canadian Plastics, letsrecycle webpage, Chemanalyst, Chemical week by S&P Global

Data

Click this link to download the data: LCC 2026 DCF